

$$T_p \left(\frac{dy}{dt} \right) + y = K_p f(t)$$

$$\mathcal{L} \left(\frac{dy}{dt} \right) = s Y(s) - y(0) \quad [y(0) = 0]$$

$$T_p s Y(s) + Y(s) = K_p F(s)$$

$$Y(s) = Y(s)(1 + T_p s) = K_p F(s)$$

$$\frac{Y(s)}{F(s)} = \frac{K_p}{T_p s + 1}$$

Transfer function.